Affine coefficient loop and bounded word arithmetic 2026-10-31

T-1142. Producer proof packet; independent source-fidelity review pending.

Checked loop model

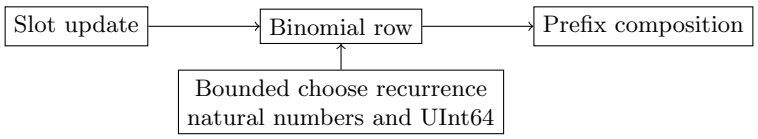
For a field, input coefficients c_k and scale s, define

$$R_k = \sum_{j=0}^{k} C\left(c_k \binom{k}{j} a^j b^{k-j} s^{-1}\right) X^j.$$

The row theorem proves $R_k = C(c_k s^{-1})(aX+b)^k$. Inner and outer prefix invariants then give, for arbitrary n,

$$\mathrm{outer}(n) = C(s^{-1}) \left(\sum_{k<n} C(c_k) X^k\right) \circ (aX+b).$$

Slot updates affect only the selected coefficient. Removing a trailing zero preserves the represented polynomial. Taking $s = a^m$ connects this loop model to the preceding T_m normalization theorem.



Word bound and controls

For every $0 \le j \le k \le 5$, the UInt64 recurrence equals$\binom{k}{j}$. Its pre-division products are at most 30, divisors are positive, and $30 < 2^{64}$. Index bounds give$j < n$, $j \le k$ and$k - j + j = k$ under the loop conditions. Degree 6 attains a product 60, so the 30 bound is deliberately limited to degree 5.

With $a = 2, b = 3, c = 4, s = 2$, rows 0/1/2 are 2, $6 + 4X$, and$18 + 24X + 8X^2$. The middle coefficient 24 rejects 12 from a missing binomial factor. Controls also cover zero/empty input, slot isolation, the final index, $\binom{5}{2} = 10$ and the attained 30 bound.

Evidence and source boundary

Thirteen main declarations and fourteen controls have separate named axiom readback. Native exits and input hashes are checked. The main attempts passed with an unused-simp warning; the first control attempt failed on mapped rational constants, then passed after correction. Controls were rebuilt against the final word theorem.

The retained source excerpt and hashes bind `genus2_model_gate.rs`:1–56. Its inverse call requires $s \neq 0$. Abstract totalized inverse at zero is not a source-execution claim. Prime-field encoding, `choose%p`, field-operation correspondence, literal array/trim/compiler behavior and geometric class transport remain separate bindings. The word theorem concerns the bounded choose recurrence.

Sources: Mathlib Taylor, evaluation/coefficient documentation, and pinned `Data/Nat/Choose/Sum.lean`. No novelty claim. No scientific campaign, curve enumeration, timing or cost ratio. Diagram shows proof dependencies; cohort and campaign limits unchanged.